

WASP-12b

R-1647

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_190214 · created 2026-08-02T07:40:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458528.6752 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.042
Transit depth (Rp/Rs)^2	0.0002 +/- 0.012 [%]
Orbital Inclination (inc)	84.41 +/- 2.88
Airmass coefficient 1 (a1)	1.78 +/- 0.16
Airmass coefficient 2 (a2)	-0.54 +/- 0.33
Scatter in the residuals of the lightcurve fit is	8.96 %
Best Comparison Star	#1 - [311, 121]
Optimal Aperture	1.86
Optimal Annulus	4.86
Transit Duration (day)	0.1145 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.042 vs 0.1178 (deviation 2.17σ)
Duration fit vs published	0.1145 d vs 0.125 d (deviation 1.24σ)
Mid-time O–C	-0.5 min
Depth SNR	0.0
Residual scatter	8.96 %

Light curve

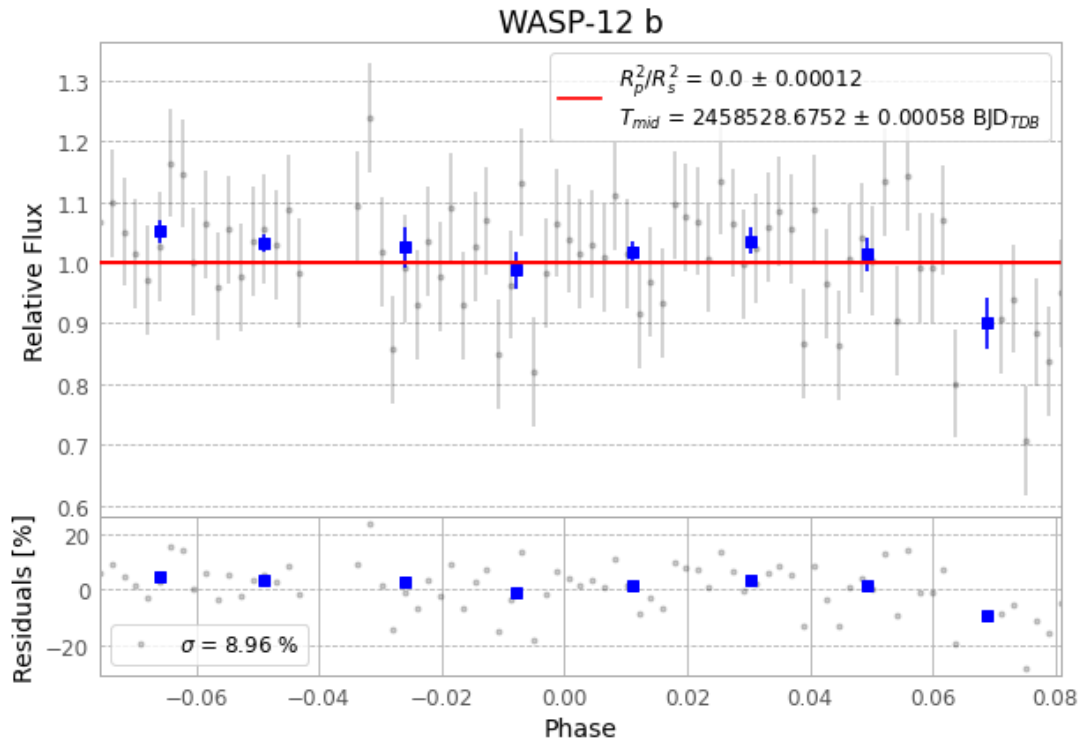


Figure 1 — FinalLightCurve_WASP-12 b_2019-02-13.png (SHA-256 70bf83f898363cec...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [424, 138], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2de3b25a64812d55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

98 FITS frames (64 MB), WASP-12190214020612.FITS → WASP-12190214065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190214.log (5d2d62a7585f...), FinalLightCurve_WASP-12 b_2019-02-13.png (70bf83f89836...), FinalLightCurve_WASP-12 b_2019-02-13.csv (6cafb373ced7...)

Compute resources

Wall time 225.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 07:40Z.